

ISEN/DAEN 427 Lecture 4

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Bayesian Inference, Estimation (cont.)

BM Ch. 4, 13



Multiple Observations: Gaussian Distributions

- Imagine measuring the length of the same pencil many times with a shaky ruler.
- Each measurement $x_1, x_2, \dots, x_N$ is a little different.
- A Gaussian distribution says: values close to the true length s are most likely, but errors can happen.
- Conditional independence: once we know the true s , each measurement is independent of the others.

Deriving the Posterior

- We want to guess the true value s using the data.
- $\text{Posterior} = (\text{Likelihood from data}) \times (\text{Our prior belief})$.
- In words:
 - Start with a belief about what s could be (the prior).
 - Update that belief with the evidence from the measurements.
- The posterior is our **updated belief**.

Estimation: ML vs MAP

- We want one best guess for s , call it $\hat{s}$.
- **Maximum Likelihood (ML):** Just guess what you measured.
Example: if you measured 10, then guess 10.
- **Maximum A Posteriori (MAP):** Mix data with prior knowledge.
 - If measurements are very reliable $\rightarrow$ trust them more.
 - If noisy $\rightarrow$ rely more on prior belief.

Estimator Quality: Bias and Variance

- How do we judge if an estimator (guessing method) is good?
 - **Bias:** Are we systematically off target?
(Always guessing a bag is 0.5 kg lighter than it really is.)
 - **Variance:** Do our guesses jump around too much?
(One time we guess 8 kg, another time 12 kg, even though the bag hasn't changed.)
- **Mean Squared Error (MSE):**

$$MSE = \text{Variance} + (\text{Bias})^2$$

- Total error comes from two sources: being consistently wrong (bias) and being too inconsistent (variance).

Example

- Suppose you want to guess the weight of a backpack by lifting it multiple times.
 - If your hand scale is always 0.5 kg too low $\rightarrow$ Bias.
 - If your lifts give wildly different results $\rightarrow$ Variance.
 - If you combine multiple lifts with a bit of prior knowledge (“a backpack usually weighs about 10 kg”), you get a more accurate estimate.

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- We assume *conditional independence*:

$$\mathbb{P}(x_1, x_2, \dots, x_N | s) = \prod_{i=1}^N \mathbb{P}(x_i | s)$$

Deriving the Posterior

- We derive the posterior distribution:

$$\mathbb{P}(s|x_1, x_2, \dots, x_N) = \kappa \prod_{i=1}^N \mathbb{P}(x_i|s)\mathbb{P}(s)$$

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- *Maximum Posterior Likelihood (MAP)*: $\hat{s}(x) = \frac{\frac{1}{\sigma^2}}{\frac{1}{\sigma^2} + \frac{1}{\sigma_s^2}} x + \frac{\frac{1}{\sigma_s^2}}{\frac{1}{\sigma^2} + \frac{1}{\sigma_s^2}} \mu_s$

Measures of Estimator Quality

- $\text{Bias}(\hat{s}|s) \triangleq \mathbb{E}[\hat{s}|s] - s$ where

$$\mathbb{E}[\hat{s}|s] = \int \hat{s}(x) \mathbb{P}(x|s) dx$$

$\triangleq$ is read as “defined as” or “is defined to be”.

- *Mean Square Error* is defined as

$$\begin{aligned} \text{MSE}(\hat{s}|s) &= \mathbb{E}[(\hat{s} - s)^2|s] \\ &= \mathbb{E}[(\hat{s} - \mathbb{E}[\hat{s}|s] + \mathbb{E}[\hat{s}|s] - s)^2|s] \\ &= \text{Var}[\hat{s}|s] + \text{Bias}(\hat{s}|s)^2 \end{aligned}$$

Bias vs Variance Decomposition

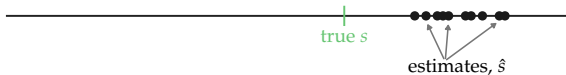
(A)

Low bias, high variance



(B)

High bias, low variance, same mean squared error



Loss Function

- **Loss function:** quantifies cost of decision.
- **Expected loss:** integrates uncertainty over θ .
- **Bayesian approach:** use posterior to minimize risk.

Loss table (state of the world)

	s_1	s_2	s_3
a_1	$L(a_1, s_1)$	$L(a_1, s_2)$	$L(a_1, s_3)$
a_2	$L(a_2, s_1)$	$L(a_2, s_2)$	$L(a_2, s_3)$
a_3	$L(a_3, s_1)$	$L(a_3, s_2)$	$L(a_3, s_3)$
$\vdots$	$\vdots$	$\vdots$	$\vdots$

Action to choose:

$$a^*(x) = \arg \min_a \sum_s L(a, s) P(y | x)$$

Note: a and s need not live in the same space

Loss Function

- A **loss function** $L(\theta, a)$ quantifies the cost of taking action a when the true parameter is θ .
- Examples:
 - Squared error: $L(\theta, a) = (\theta - a)^2$
 - Absolute error: $L(\theta, a) = |\theta - a|$
- Choice of L encodes preferences and consequences of decisions.

Bayes Rule (After Seeing Data)

Bayes action/estimator:

$$a^*(x) \in \arg \min_{a \in \mathcal{A}} \mathcal{L}(a \mid x) = \arg \min_a \mathbb{E}_{\theta|x}[L(a, \theta)]$$

The optimal decision $a^*(x)$ given data x is the action a that minimizes the expected loss, where the expectation is taken over the uncertainty about the parameter θ after observing x .

Common losses and their Bayes rules:

- *Squared error*: $L(a, \theta) = (a - \theta)^2 \Rightarrow a^*(x) = \mathbb{E}[\theta \mid x]$ (posterior mean).
- *Absolute error*: $L(a, \theta) = |a - \theta| \Rightarrow a^*(x) = \text{median}(\theta \mid x)$ (posterior median).
- *0-1 loss*: $L(a, \theta) = \mathbf{1}\{a \neq \theta\} \Rightarrow a^*(x) = \text{mode}(\theta \mid x)$ (MAP) for discrete Θ .

- Since θ is unknown, we consider the **expected loss**.
- Frequentist view: expectation w.r.t. sampling distribution of data.
- Bayesian view: expectation w.r.t. posterior distribution:

$$\rho(a | x) = \mathbb{E}[L(\theta, a) | x] = \int L(\theta, a) p(\theta | x) d\theta$$

- Goal: choose $a^*(x) = \arg \min_a \rho(a | x)$

Bayesian Decision Rule

- In the Bayesian framework:
 - 1 Prior $p(\theta)$ encodes beliefs about parameters.
 - 2 Posterior $p(\theta \mid x)$ updates beliefs after observing data.
 - 3 Decision rule: minimize posterior expected loss.
- Examples:
 - With squared-error loss, the Bayes estimator is the posterior mean:
$$a^*(x) = \mathbb{E}[\theta \mid x]$$
 - With absolute-error loss, the Bayes estimator is the posterior median.

Loss Function

- Imagine making a guess about something unknown, like the bias of a coin.
- A **loss function** tells us how bad our guess is if we are wrong.
- Example: If the true answer is 0.7 and we guess 0.6, our loss might be $|0.7 - 0.6| = 0.1$

Why Why Why

- Without a loss function, we don't know which guesses are better.
- It gives us a **rule** to decide.
- Different loss functions lead to different decisions!

Bayesian Idea

- We start with a belief (called a **prior**).
- We collect data (like coin flips).
- We update our belief to a **posterior**.
- Then, we pick the guess that makes the **expected loss** smallest.

Example: Coin Flip

- Suppose we flip a coin 3 times and see 2 heads.
- Prior belief: the coin could be fair (50-50).
- After data: it looks more likely the coin is slightly biased towards heads.

Loss function: squared error $L(\text{guess}, \theta) = (\text{guess} - \theta)^2$

- Best guess = the **average** of the posterior (the mean).

Numerical Example

- Prior: think of coin bias as $\text{Beta}(1,1)$ = uniform (all values 0–1 equally likely).
- Data: 2 heads, 1 tail.
- Posterior: $\text{Beta}(3,2)$.
- Posterior mean = $\frac{3}{3+2} = 0.6$.

So our best guess is 0.6 under squared error loss.

- Loss functions measure how wrong a guess is.
- Bayesian inference updates beliefs with data.
- The best guess depends on the loss function!